Indian Institute of Science Education and Research Kolkata

Department of Mathematics and Statistics

Subject: Probability and Statistics-I (MA2104)

End-semester Examination Full marks: 50

Date of Examination : 05.12.2025 Max. time: 150 minutes

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Instructions: There are five questions in all. Answer all the questions. In order to get full credit, all the intermediate steps must be provided and the answers must be given in its most simplified forms. Every answer must be explained with proper reasoning. Calculator may be allowed. However, exchange of calculator is not permitted. The notations have their usual meanings. The number inside the square brackets at the right of each question denotes the marks corresponding to that question.

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1. (a) Let a random variable X be uniformly distributed over $\{1, 2, \ldots, 50\}$. Evaluate the following probability. [5]

$$P\left(X + \frac{96}{X} > 50\right).$$

(b) Given that $P(A) = 3/4$ and $P(B) = 1/3$. Prove that [5]

$$\frac{1}{12} \leqslant P(A \cap B) \leqslant \frac{1}{3}.$$

2. Answer the following. [5× 2=10]

(a) Suppose X follows the symmetric binomial distribution with $n = 5$. Find the probability of at least 4 successes.

(b) It is known that a binomial distribution attains its maximum possible variance. If the distribution has $n = 6$, find $E(X^2)$.

(c) Let X be a positive random variable with $E(X) = 2$ and $V(X) = 3$. If $E(X^3) < \infty$, show that $E(X^3) > 18$.

(d) For two random variables X and Y, show that $V(E(X|Y)) = 0$ if and only if $E(X|Y = y)$ is independent of y.

(e) Show that if X and Y are two identically distributed random variables, then $X - Y$ may not be symmetric.

3. Identify which of the following statements may be correct and which are not. Justify your answers. [5×2=10]

(a) Standard deviation of X is 2.8, mean deviation about mean of X is 3.2

(b) For a data set $(x_1, x_2, \ldots, x_{10})$, median of x is 5, $\sum_{i=1}^{10} x_i = 40$, $\sum_{i=1}^{10} |x_i - 5| = 25$, $\sum_{i=1}^{10} |x_i - 4| = 27$.

(c) Mean of X is 8, median of X is 4 and variance of X is 3.24

(d) If the range of a variable is 6, its variance cannot be more than 9.

(e) Suppose X and Y are two independent negative random variables. Then $E(XY)$ is negative.

4. (a) Suppose the deviations $x_i - \bar{x}$ are small compared to $\bar{x}$ so that $\left(\frac{x_i-\bar{x}}{\bar{x}}\right)^3$ and higher powers of $\left(\frac{x_i-\bar{x}}{\bar{x}}\right)$ may be neglected. Prove that [5]

$$\left(\prod_{i=1}^{n} x_i^{1/2}\right)^{1/n} \approx \sqrt{\bar{x}}\left(1 - \frac{s^2}{4(\bar{x})^2}\right).$$

(b) For a dataset $(x_1, x_2, \ldots, x_n)$, show that the mean deviation about median is the least. [5]

5. Calculate (i) Arithmetic mean, (ii) Median and (iii) Mode from the following frequency distribution. [10]

Classes	Frequency
0-4	5
4-8	12
8-12	13
12-16	7
16-20	3

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